Definitions:

Definition 1.1: Let A and B be two sets. We say that A is a subset of B, or that A is contained in B if and only if each x $\in$A is also an element of B. We write A $\subseteq$ B to say that "A is a subset of B." If there is an element of A not in B, we write A $\nsubseteq$ B.

Definition 1.2: The empty set, written $\emptyset$, is the set that has no elements.

Definition 1.3: If A and B are statements, their conjunction is the statement "A and B". A $\wedge$ B is true exactly when both A is true and B is true

Definition 1.4: The intersection S$\cap$T is the set consisting of the elements S and T have in common. One can write S$\cap$T=$x : (x \in S) \wedge (x \in T)$

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